CURRICULUM VITAE · SEPTEMBER 2026

Wei-Tao Lyu

Postdoctoral Fellow in Physics · Chinese University of Hong Kong

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RESEARCH INTERESTS

- Microwave kinetic inductance detectors
- Submillimetre astronomy
- Superconducting resonators
- Cryogenic instrumentation
- Detector readout and calibration

TECHNICAL SKILLS

Instrumentation
Superconducting resonators, MKIDs, device characterization

Cryogenics
Cryostat integration and optimization for astronomy systems

Data analysis
Signal processing and calibration for superconducting devices

Software
Simulation and data analysis tools for instrumentation work

PROFILE

Astronomical instrumentation researcher working on superconducting MKIDs, submillimetre camera systems, cryogenic integration, and readout characterization.

APPOINTMENTS

2022–present — **Postdoctoral Fellow**
Department of Physics, Chinese University of Hong Kong
Supervisor: Prof. Hua-Bai Li

2022 — **Research Associate**
Department of Physics, Chinese University of Hong Kong
Supervisor: Prof. Hua-Bai Li

EDUCATION

2015–2021 — **PhD in Astronomy**
School of Astronomy and Space Science, University of Science and Technology of China
Advisor: Prof. Sheng-Cai Shi

2011–2015 — **Bachelor of Science in Physics**
School of Physics, Huazhong University of Science and Technology

SELECTED PUBLICATIONS

1. **Laser-Based Frequency-to-Pixel Mapping Method for Submillimeter MKID Array.** Lyu, W., Li, H. B., Zhi, Q., Li, J., & Shi, S. (2026). *RAS Techniques and Instruments*, rzag007.
2. **□□□□□□□□□□Roger.** □□□□□□□□□□□□□□□ (2025). □□43 □□4□.
3. **ROGer: Remote Observing from Greenland.** Weitao Lyu, Junkun Huang & Hua-bai Li (2024). *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, 131021D.
4. **Disorder effects in NbTiN superconducting resonators.** Lyu, W. T., Zhi, Q., Hu, J., Li, J., & Shi, S. C. (2024). *Chinese Physics B*, 33(2), 027401.
5. **Cavity and ground effects on a superconducting microstrip resonator.** Lv, W. T., Ding, J. Q., Zhi, Q., Wang, Z., Li, J., & Shi, S. C. (2021). *AIP Advances*, 11(9), 095308.

RESEARCH PROJECT

Hong Kong's Window on the Universe
Building a pioneering submillimetre astronomical camera. Role: cryostat system integration with superconducting MKIDs and cold optics.